



Handouts 0

Operating Systems

No. 0

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What is Computer's Operating System?

the software that manages a computer's resources.

What will we learn?

How an operating system works?
How to write a program that uses the operating system functions?

Why do we need to learn?

You may need to develop an OS of your own.

The core ideas in a modern operating system — protection, concurrency, virtualization, resource allocation, and reliable storage — have become widely applied throughout computer science

Why do we need to learn?

In a modern world, nearly everything a user does is distributed, nearly every computer is multi-core, security threats abound, and many applications such as web browsers have become mini-operating systems in their own right.

Why we need to learn?

```
int a[1024][1024];
// We need to initialize array a, we can write the code as follow
for (int i = 0; i < 1024; i++) {
    for (int j = 0; j < 1024; j++) {
        a[i][j] = 0;
    }
}
// and as follow
for (int i = 0; i < 1024; i++) {
    for (int j = 0; j < 1024; j++) {
        a[i][j] = 0;
    }
}
Which one is better?
```

Why do we need to learn?

Become a power user.
Understand the benefit of using command line.

Programming Languages

C
Java

OS we will work with Ubuntu http://www.ubuntu.com/download	Linux installation using VM concept Download Oracle VirtualBox https://www.virtualbox.org/ Install the program Create a virtual machine Install Ubuntu on the created virtual machine
The way we work with Ubuntu in this course However, in our class we will use only command line interface. So we may not need to install the entire OS.	The way we work with Ubuntu in this course For Windows Enable WSL (Windows Subsystem for Linux) For Mac Use terminal (some command and some programs may not 100% compatible)

Notes

✓ What is Computer's Operating System (OS)?

- the software that manage computer resources.

✓ What will we learn?

because you used it everyday and os today is pretty easy to use, in our class, we don't learn how to use os, we'll learn

- How an operating system works?
- How to write a program that uses the operating system functions?

✓ Why do we need to learn?

1. You may need to develop an OS of your own.
 - Actually, OS is a very complex software and of course you can use your software engineering knowledge apply to OS development. But before that, you have to understand how os work, what function os should provide for us.

some of you may argue, I won't go out to be os developer.

The core ideas in a modern operating system — protection, concurrency, virtualization, resource allocation, and reliable storage — have become widely applied throughout computer science.

the core idea that in the past it just apply to os, now they apply to computer science in general (almost all program that you gonna going out to develop still need to support this concepts)

Example



In a modern world, nearly everything a user does is distributed, nearly every computer is multi-core, security threats abound, and many applications such as web browsers have become mini-operating systems in their own right.

- **Distributed** — we use application like the cloud application we have to connect to network, this mean there more than 1 programs running at the same time to helps us finish our task. the program may need to share resources (resource allocation, concurrency control)
- **Multi-core**, — even tho, we only focus on single device maybe computer or mobile device like mobile phone or tablet — almost all of them doesn't have single core cpu, they have multi-core cpu. → this mean even in 1 device, we have to think about concurrency control.
- **Security threats abound** — because we connect to the network, mean that we have to consider more about security about protection, how we going to protect device from security threats
- **Many applications such as web browsers have become mini-operating systems in their own right** — because as you can see, we can open many tab under web browser, behind those tab, there are program running maybe javascript program running. — in the sense that it need to do the resources allocation for each program, it need to protect the programs, don't interfere to each other. (this is the example of the task of os)

2. You may become a better programmer.

Which one is better?

```
int a[1024][1024]
//We need to initialize array a, we can write the code as follows (assign all positions to value 0)
```

- First way

```
for (int i = 0; i < 1024; i++) {
    for (int j = 0; j < 1024; j++) {
        a[i][j] = 0;
    }
}
```

- Second way

```
for (int j = 0; j < 1024; j++) {
    for (int i = 0; i < 1024; i++) {
        a[i][j] = 0;
    }
}
```

Based on what you learn in C programming course — we use the first one, but why? you'll know the answer after you finish this course

3. Become a power user — you'll understand the idea behind the scene of how OS works, it may help you be like if I use this way, it's better.
4. Understand the benefit of using command line. — because in our class, we'll use command line to control the system — you may say why I have to use command line, GUI is easy to use.
 - let's say you want to delete 1000 of doc files from folder

| In addition, some tasks you can't use GUI.

| command line is also essential for many tech jobs.

✓ Programming languages

- C - we gonna write a program to communicate to the os, and C is a system programming language that can be use for this purpose
- Java - we will focus on concurrency features of java

We gonna learn features provided by java and compare to C

✓ OS we will work with

- Ubuntu
 - <http://www.ubuntu.com/download>

If you want to install Ubuntu and use it directly, there are 2 major option

1. You create a new partition on your hard drive or your ssd drive and install ubuntu on that partition — not suggest for less-experience user
2. Install ubuntu on your external ssd drive
3. (Easier way, enough ram) - use virtual machine
 - download oracle virtual box
 - install the program
 - create virtual machine
 - install ubuntu on the created virtual machine

✓ The way we work with Ubuntu in this course

- However, in our class we will use only command line interface
- So, we may not need to install the entire os.

For Windows

- enable WSL (window subsystem for linux)

For Mac (Unix-based)

- use terminal (some commands and some programs may not 100% compatible)
- to do - make sure your system has gcc compiler - install from Xcode

